DeepSeek Sparse Attention (DSA)

A lightning indexer picks 2,048 keys per query

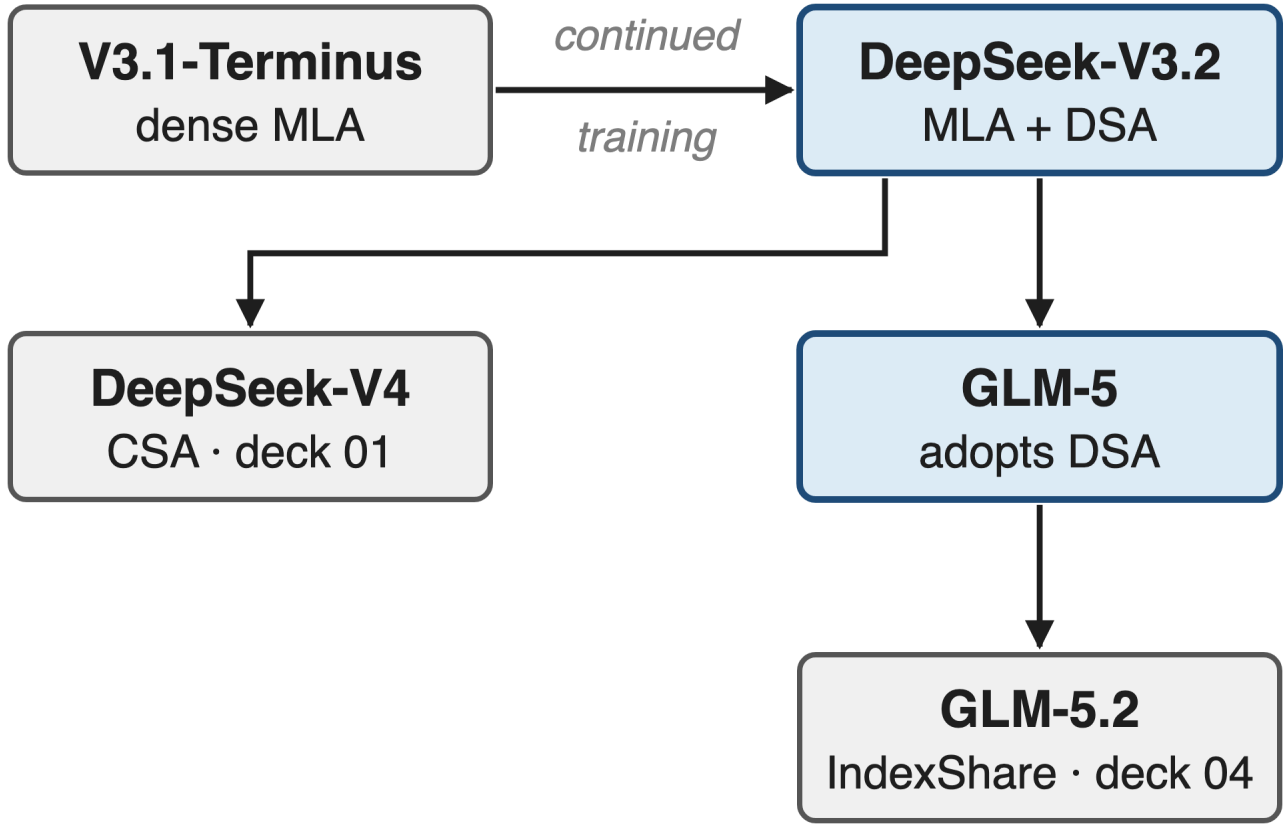
DeepSeek-AI · DeepSeek-V3.2 (adopted by GLM-5)

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Attention & Long Context · arxiv.org/abs/2512.02556

DSA is V3.2's only architectural change

- V3 backbone: **671B** MoE, **37B** active, MLA
- **128K** context, continued from V3.1-Terminus
- Adopted by GLM-5 (**744B**)



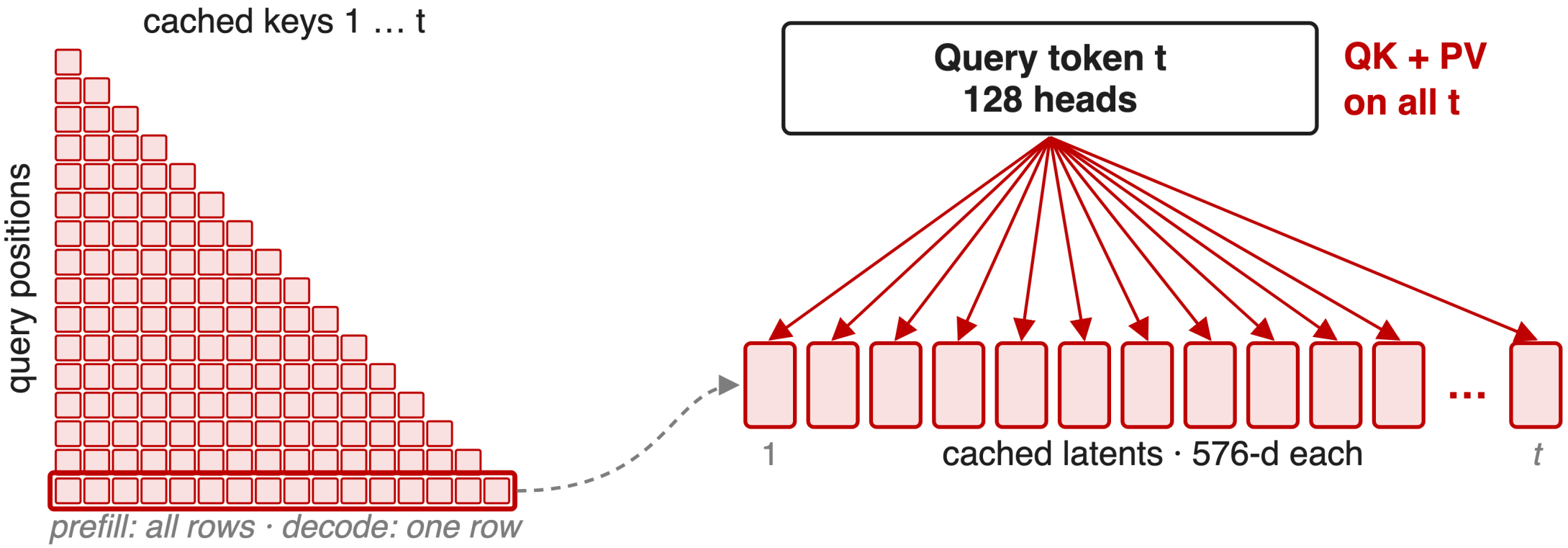
Dense attention makes late tokens 31× costlier

31×

decode cost per token at position 128K vs position 0 with dense MLA: ≈ $2.15 vs ≈ $0.07 per million tokens

DeepSeek-V3.2 report, Fig. 3(b): H800 service at $2/GPU-hour; values read from the figure's vector data

Every token's attention reads every cached latent



Villain: the 128-head attention itself decides which keys matter, so every key costs 139,264 multiply-adds

Fixed patterns and linear attention lose recall

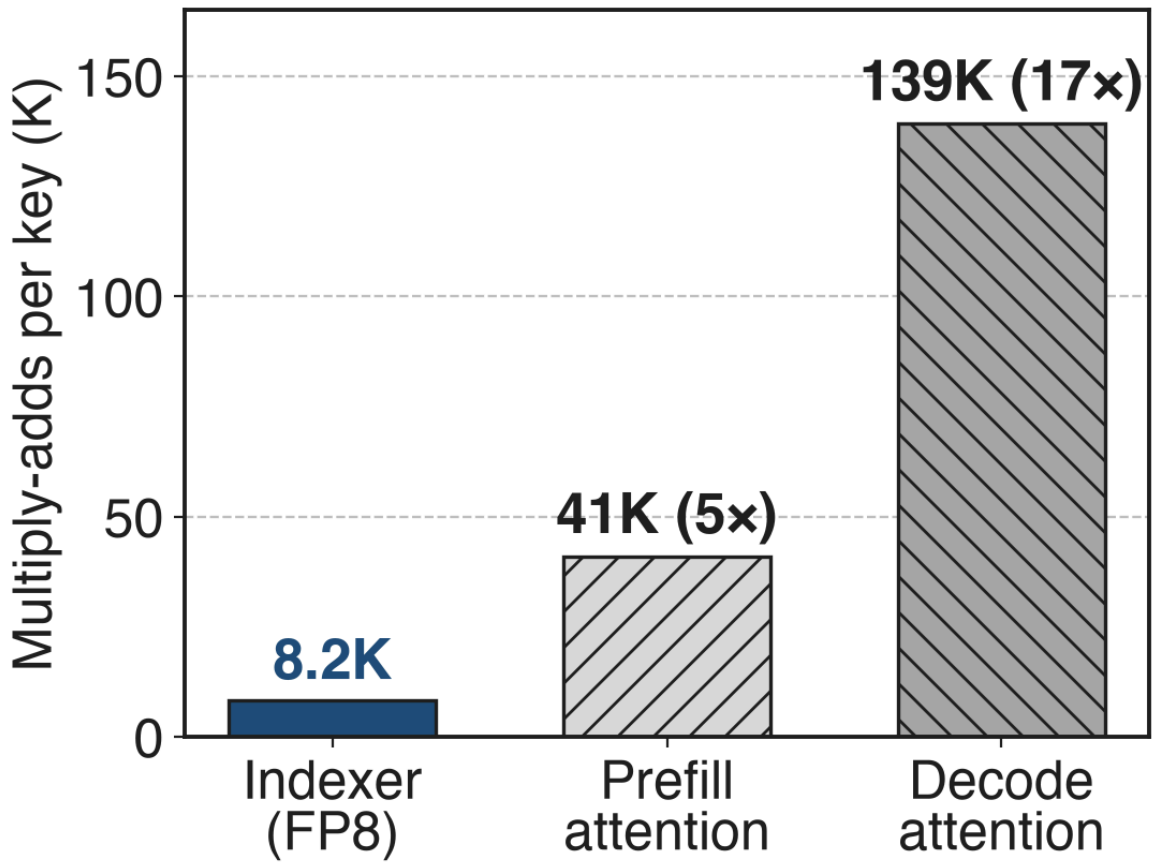
- **Fixed windows** (SWA, half the layers full)
- RULER@128K: **–30.4** interleaved, **–5.7** searched

- **Linear attention** (GDN variants)
- RULER@128K: **–11.3** / **–8.3**
- Native sparse: **new pretraining run**

Needed: content-based selection, added to an already-trained dense model

Insight: scoring is 17× cheaper than attending

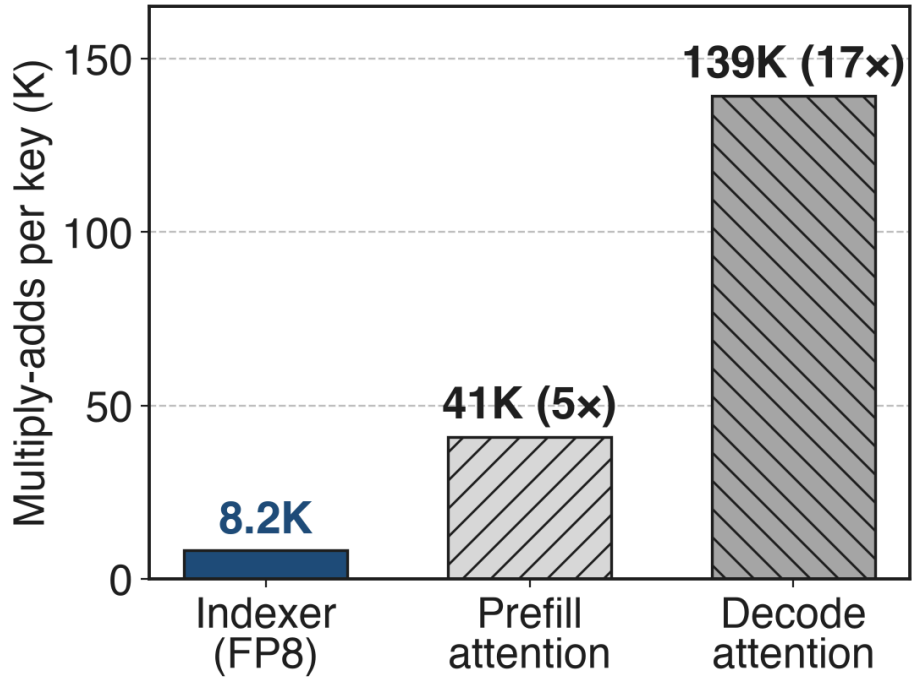
- Score a key: **64 × 128** FP8 multiply-adds
- Attend to it: **128 × 1,088** BF16 (decode)



Analytic, per query per layer, V3.2 shapes

Insight: scoring is 17× cheaper than attending

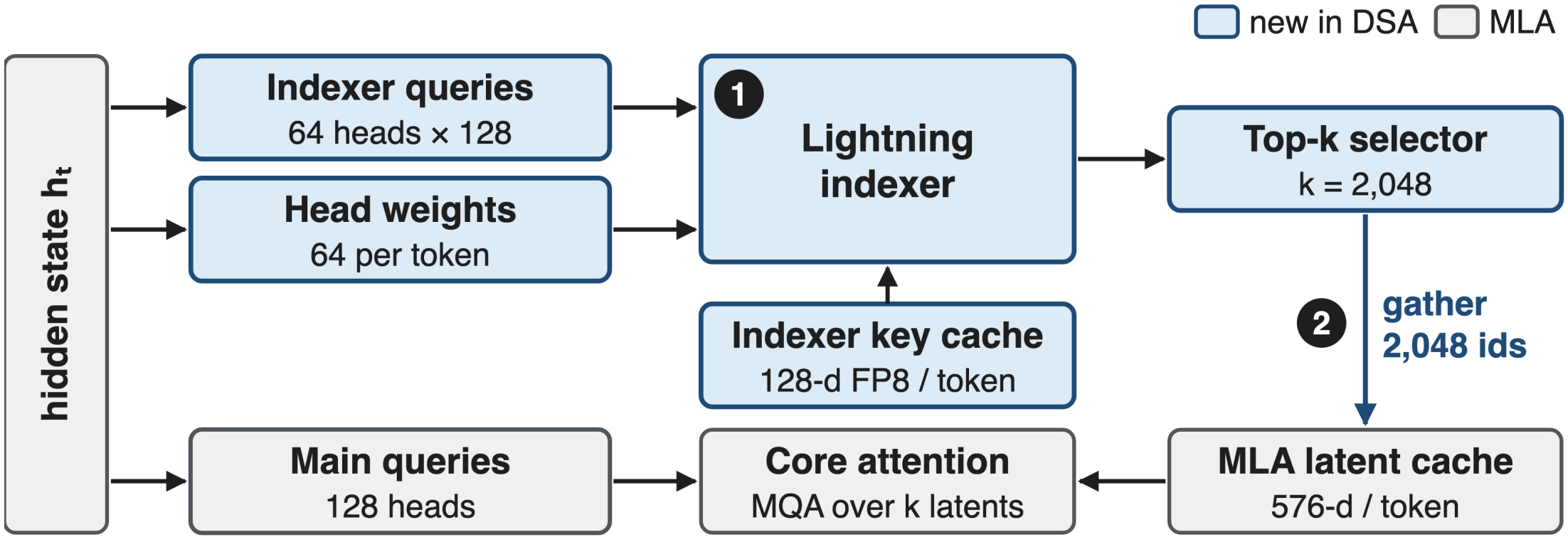
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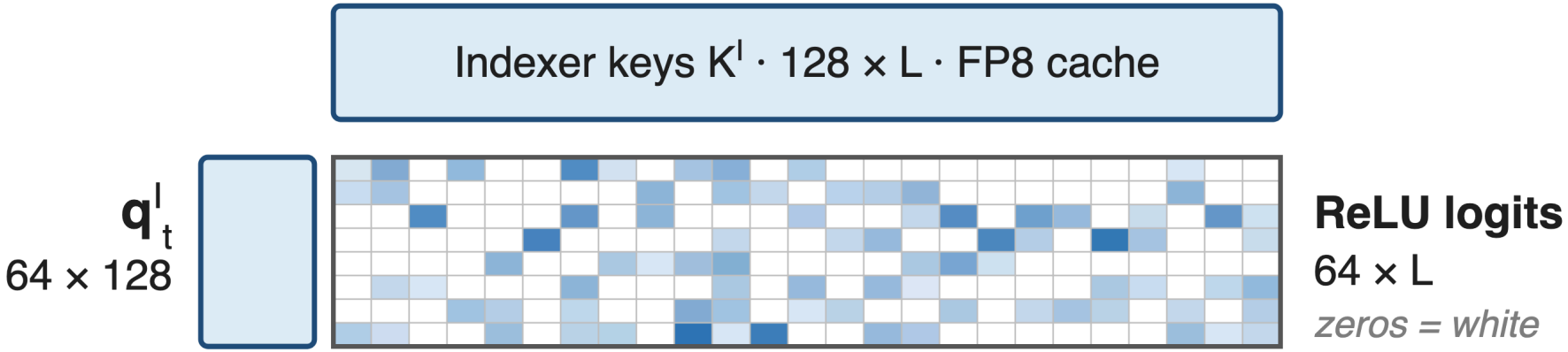
Let a tiny FP8 scorer read every key; attention reads only the top 2,048 (1.6% of 128K)

DSA puts a cheap scorer in front of MLA



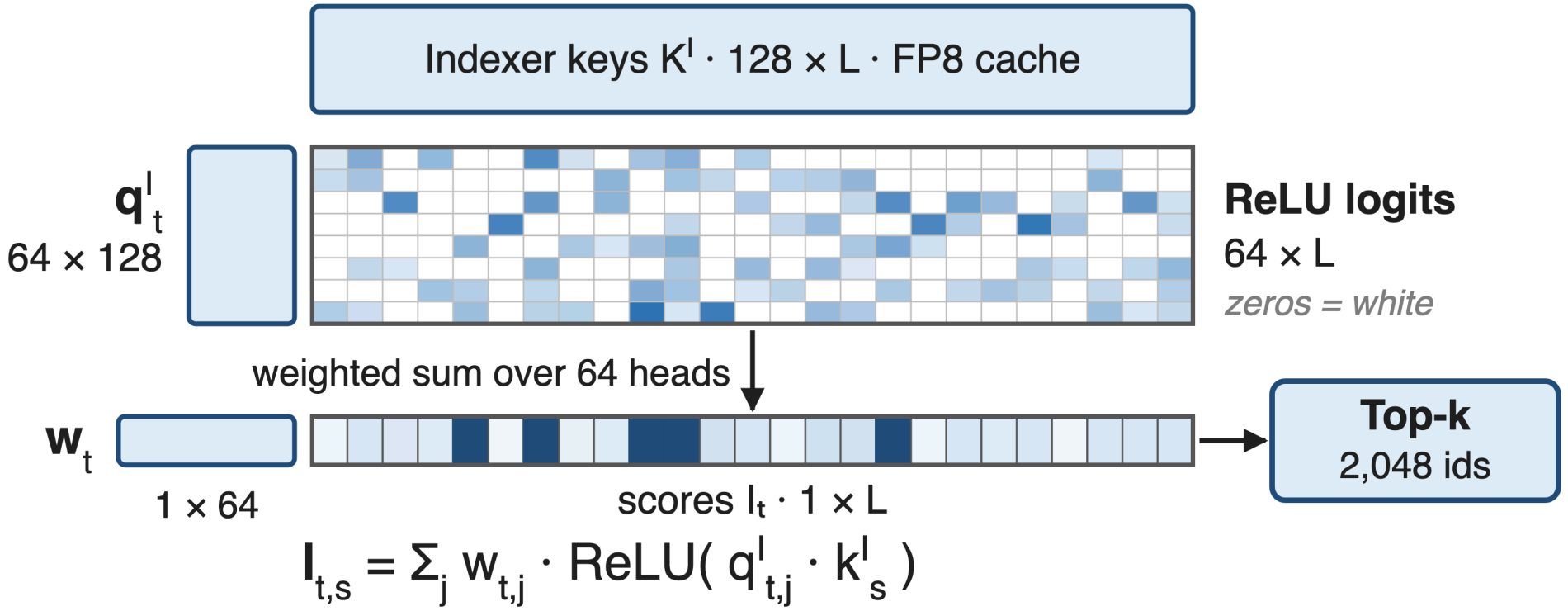
❶ the indexer scores every cached token · ❷ the top 2,048 latents feed MQA-mode attention · ❸ continued training (not drawn)

❶ The lightning indexer scores every key in FP8



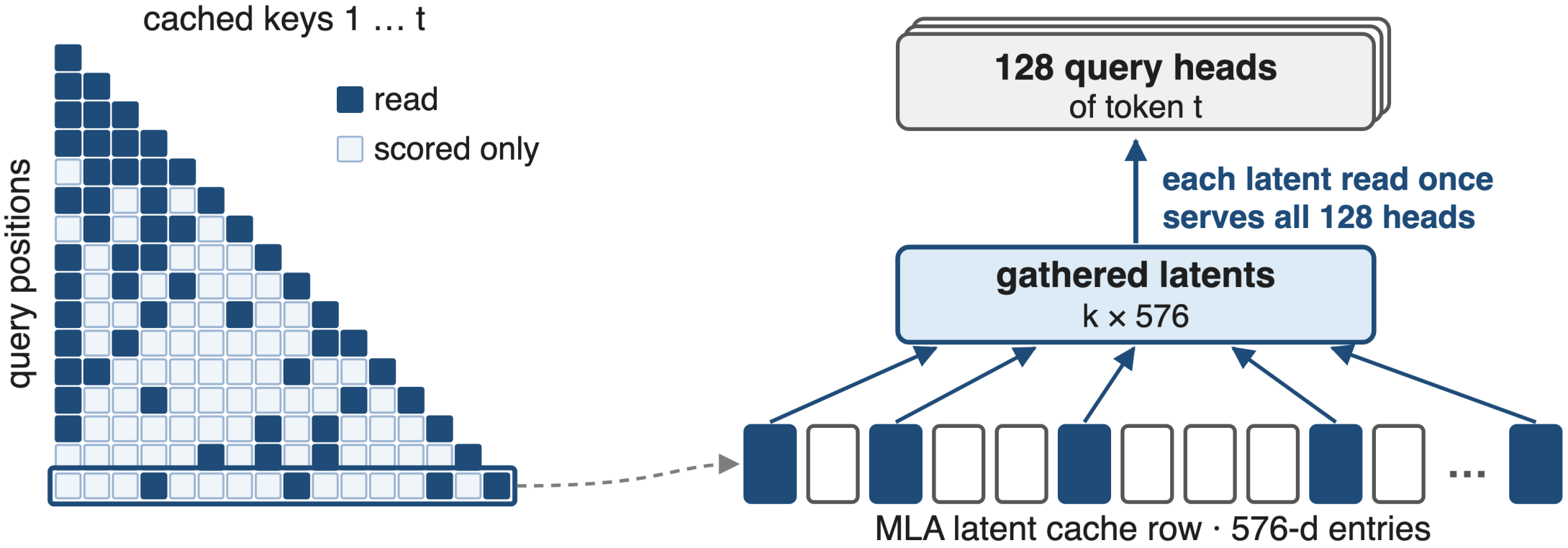
$$I_{t,s} = \Sigma_j \, w_{t,j} \cdot \mathrm{ReLU}(q^I_{t,j} \cdot k^I_s)$$

❶ The lightning indexer scores every key in FP8



ReLU instead of softmax, one 128-d key per token shared by all 64 heads, E4M3 inputs

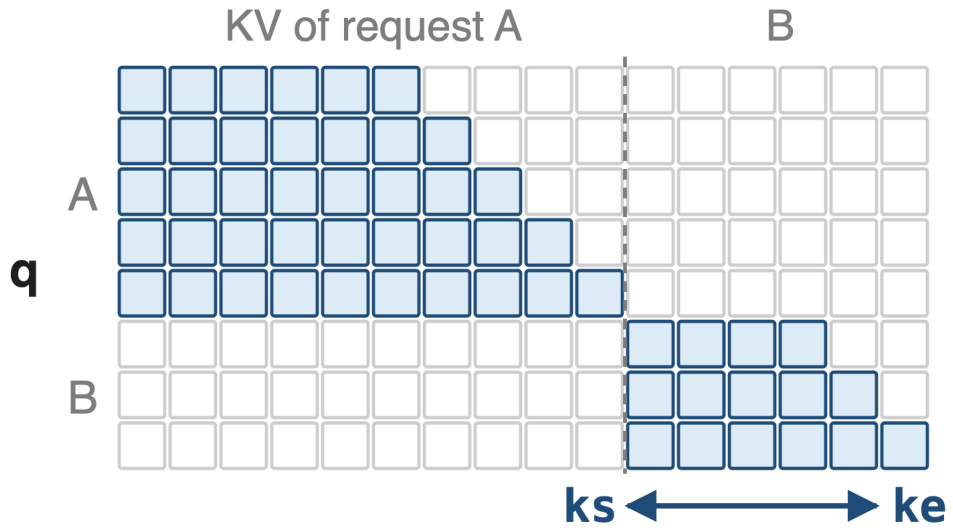
❷ 2,048 shared latents serve all 128 heads



MQA mode: a token's 576-d latent is the key (512-d part the value) for all 128 heads

Prefill and decode call different indexer kernels

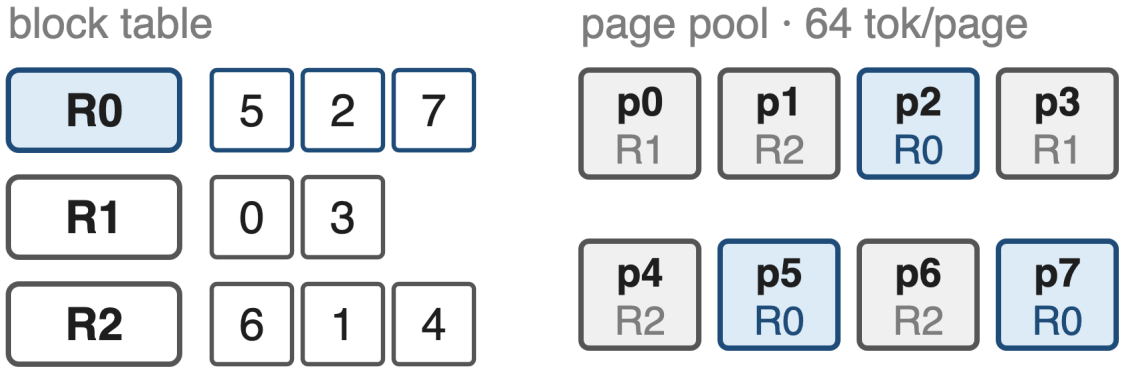
(a) Prefill: packed requests



one [ks, ke) range per query row

`fp8_mqa_logits`

(b) Decode: paged KV cache



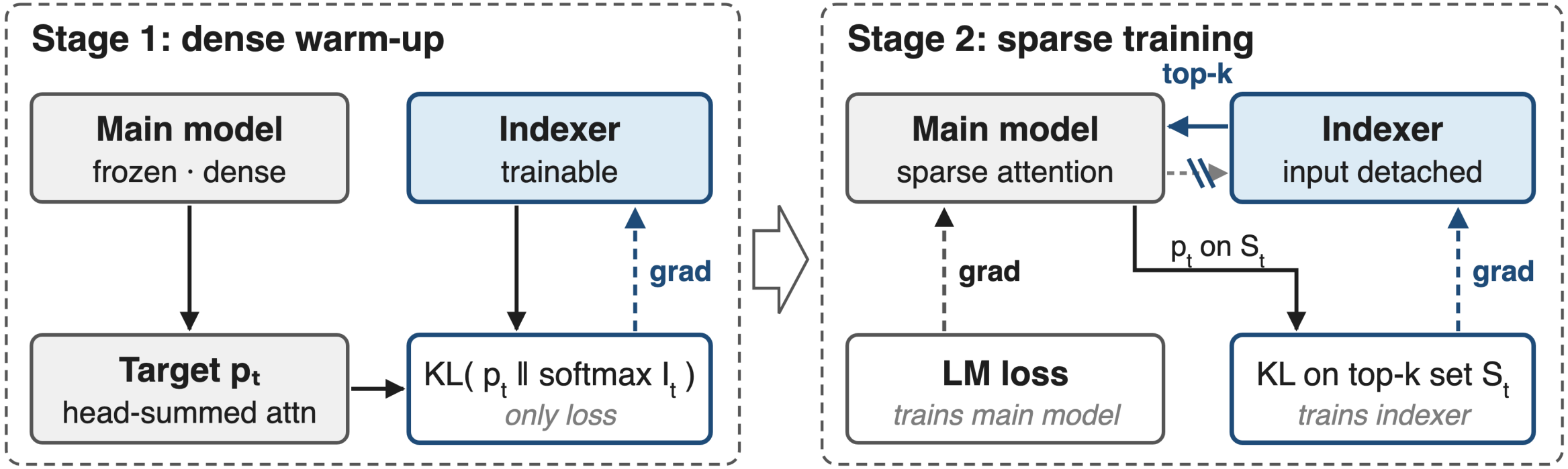
new token(s) of each request

pages located via block table

`fp8_paged_mqa_logits`

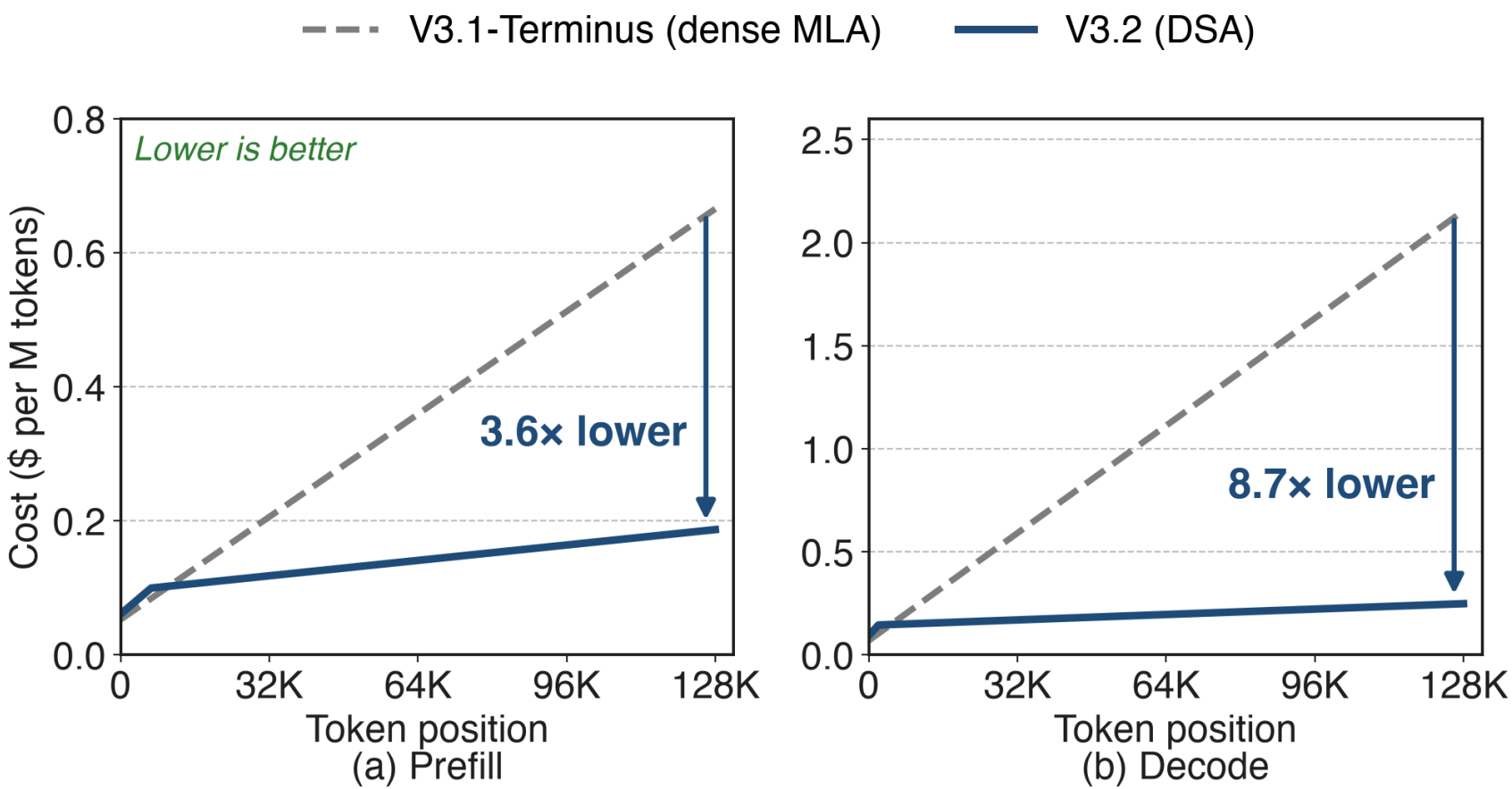
Both emit one score row per query, then top-2,048 feeds FlashMLA's sparse kernel (kernels: deck 10)

❸ Warm up the indexer, then train sparse



Warm-up: 1,000 steps, 2.1B tokens · sparse stage: 15,000 steps, 943.7B tokens of 128K sequences

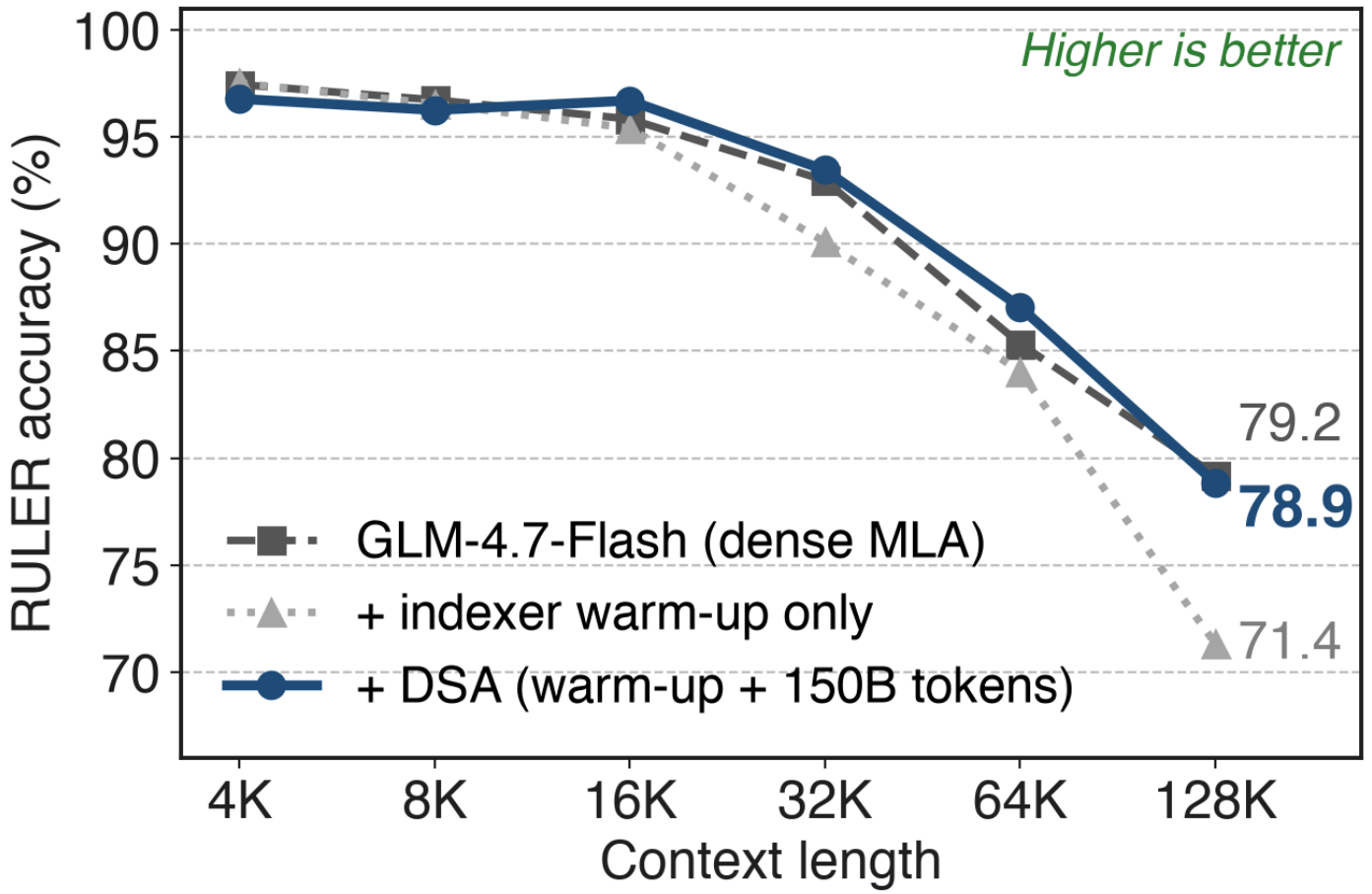
DSA cuts 128K decode cost 8.7×



H800 service, $2/GPU-hour; curves read from DeepSeek-V3.2 Fig. 3. Below ≈ 5K (decode) / ≈ 10K (prefill) tokens DSA costs slightly more.

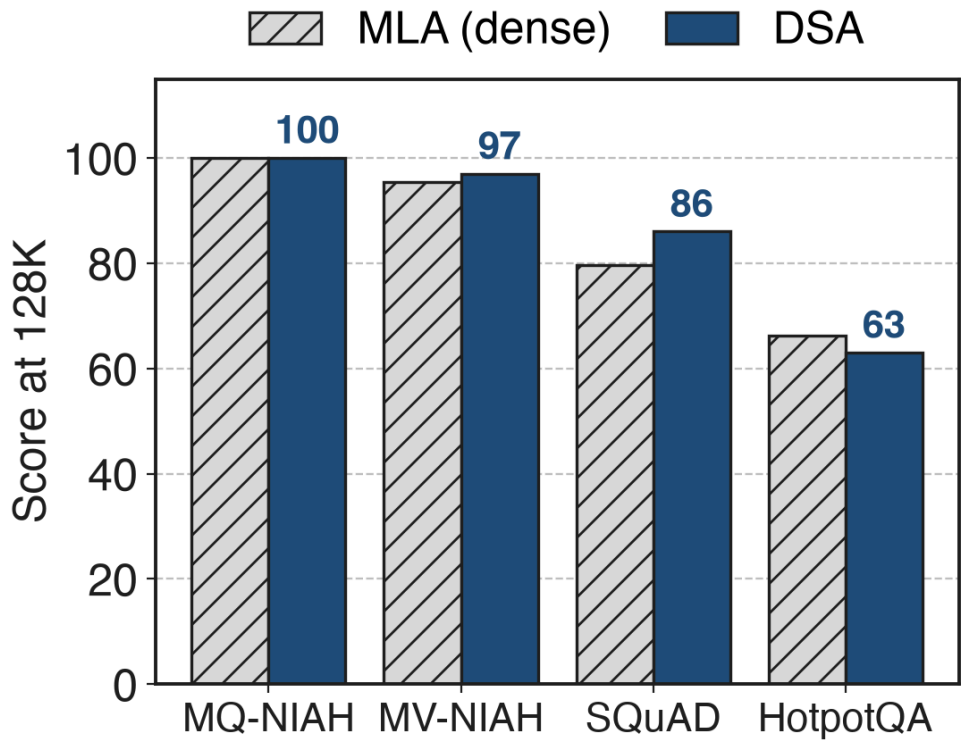
Quality holds: 128K RULER within 0.35 points

- V3.2-Exp: **+4** on AA-LCR vs V3.1-Terminus
- Warm-up alone: **–7.9** at 128K
- Joint training: **–0.35** at 128K



GLM-5 adopts DSA with 47× fewer tokens

- Sparse adaptation: **20B** tokens vs DeepSeek's **943.7B**
- **32** indexer heads (V3.2: **64**)
- RL: deterministic `torch.topk`, frozen indexer



Non-deterministic top-k kernels wrecked RL within a few steps

The cost: an $O(L^2)$ indexer and new state

- **Indexer still $O(L^2)$**: **79%** of per-query work at 128K
- Short contexts: slightly costlier
- +**132 B** key cache per token per layer (+20%)
- Needs continued pretraining

Next: share one indexer across layers (deck 04), make it fast (deck 10)

Summary

- Dense MLA: decode cost per token grows **31×** from 0 to 128K
- **Insight**: scoring a key is **17×** cheaper than attending to it
- FP8 indexer → top-2,048 → MQA-mode sparse MLA, via continued training
- At 128K: **8.7×** lower decode, **3.6×** lower prefill cost; quality on par

https://arxiv.org/abs/2512.02556